3.5 Summary

- Prior knowledge plays an important role in a Bayesian model and can heavily influence the posterior predictions of the model.
- With a GP, prior knowledge can be specified using the mean and the covariance functions.
- The mean function describes the expected behavior of the GP model. In the absence of data, a GP's posterior mean prediction reverts back to the prior mean.
- A GP's mean function can take on any functional form, including a constant, a linear function, and a quadratic function, which can be implemented with GPyTorch.
- A GP's covariance function controls the smoothness of the GP model.
- The length scale specifies the level of variability of the input with respect to the function output. A large length scale leads to more smoothness and thus less uncertainty in the predictions.
- Each dimension in a GP can have its own length scale. This is called automatic relevance determination, or ARD, and is used to model objective functions that have different levels of variability in different dimensions.
- The output scale specifies the range of the function output. A large output scale leads to a larger output range and thus more uncertainty in the predictions.
- The Matérn kernel class is a generalization of the RBF kernel class. By specifying its parameter `nu`, we can model various levels of smoothness in the GP's predictions.
- The hyperparameters of a GP can be optimized by maximizing the marginal likelihood of the data using gradient descent.

3.6 Exercise

This exercise provides a practice for implementing a GP model with ARD (automatic relevance determination). To do this, we will create an objective function that varies along an axis more than it does along another axis. We will then train a GP model, with and without ARD, on data points from this function and compare the learned length scale values.

There are multiple steps:

1. Implement the following two-dimensional function in Python using PyTorch:

$$[f(x_1, x_2) = \sin\left(\frac{5x_1 - 5}{2}\right)\cos(2.5 - 5x_2) + \frac{1}{10}\left(\frac{5x_2 + 1}{2}\right)^2]$$

This function simulates the accuracy surface of a support-vector machine (SVM) model in a hyperparameter tuning task. The x-axis denotes the value of the penalty parameter c, while the y-axis denotes the value for the RBF kernel parameter γ. (We will be using this function as our objective in future chapters as well.) . Visualize the function over the domain `[0, 2]` ^{2}. The heat map should look like Figure 3.17.

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